

Chapter 11: Protozoan Groups

UNICELLULAR EUKARYOTES

Emergence of Eukaryotes and a New Life Pattern

- First evidence of life dates to 3.5 billion years ago (prokaryotic bacteria-like organisms).
- Eukaryotic cells evolved from prokaryotic ancestors via **symbiogenesis** (engulfment without digestion).
- **Mitochondria** originated from aerobic prokaryotes; **plastids** originated from photosynthetic bacteria (**primary endosymbiosis**).
- **Secondary endosymbiosis** occurs when a eukaryotic cell obtains plastids from another eukaryote.
- **Protozoa** are unicellular eukaryotes with animal-like features (absence of cell wall, presence of motile stage).
- Protozoa are complete organisms where all life activities occur within a single plasma membrane.
- They are highly adaptable, found everywhere life exists, and play a huge role in the economy of nature (e.g., ocean soil deposits, symbiosis, parasitism).

HOW DO WE DEFINE PROTOZOAN GROUPS?

- Protozoa were traditionally a single phylum but are not monophyletic; they belong to multiple eukaryotic clades.
- **Opisthokonta** includes choanoflagellates, animals, and fungi.
- **Viridiplantae** includes green algae and plants.
- **Protist** is a general term for unicellular eukaryotes; **protozoan** usually refers to animal-like unicells.



Figure 1: Cladogram showing two major prokaryotic branches and diversification of eukaryotes. Major eukaryotic clades containing protists are shown, but several clades of amebas and other forms are not shown. The order of branching remains to be determined for most clades. The very large opisthokont clade contains choanoflagellates, fungi, and all multicellular animals.

- The cladogram shows the diversification of eukaryotes into major clades like Opisthokonta, Amoebozoa, Viridiplantae, etc.

Nutrition

- **Autotrophic:** Synthesize own organic constituents (e.g., photosynthesis).
- **Heterotrophic:** Obtain organic molecules synthesized by others.
 - **Phagotrophs** (holozoic): Ingest visible particles via **phagocytosis**.
 - **Osmotrophs** (saprozoic): Ingest soluble food.
- Nutritional modes are variable and opportunistic (e.g., **Euglena** can be autotrophic or saprozoic).



Figure 2: Some feeding methods among protozoa. **Amoeba** surrounds a small flagellate with pseudopodia. **Leidyopsis**, a flagellate living in the intestine of termites, forms pseudopodia and ingests wood chips. **Didinium**, a ciliate, feeds only on **Paramecium**, which it swallows through a temporary cytostome in its anterior end. **Podophrya** is a suctorian ciliophoran. Its tentacles attach to its prey and suck prey cytoplasm into the body of the **Podophrya**, where it is pinched off to form food vacuoles. **Codonosiga**, a sessile flagellate with a collar of microvilli, feeds on particles suspended in the water drawn through its collar by the beat of its flagellum.

- Various feeding methods include pseudopodia (**Amoeba**), temporary cytostomes (**Didinium**), tentacles (**Podophrya**), and collar microvilli (**Codonosiga**).

Locomotion

- Traditional classification used locomotion: flagellates, ciliates, and amebas.
- **Flagella** and **cilia** (undulipodia) are morphologically similar but differ in propulsion mechanics.
 - Cilia propel water parallel to the surface.
 - Flagella propel water parallel to the main axis.

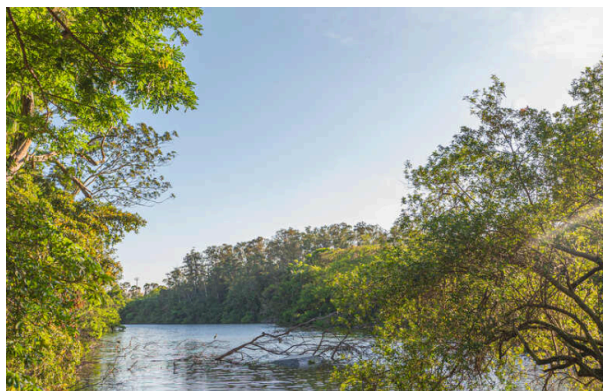


Figure 3: One flagellum is clearly visible in the lower left of this photograph of **Euglena**.

- **Euglena** moves using a long flagellum.



Figure 4: Scanning electron micrograph of a free-living ciliate **Tetrahymena thermophila** showing rows of cilia (2000x). Beating of flagella either pushes or pulls the organism through its medium, while cilia propel the organism by a “rowing” mechanism.

- **Tetrahymena** moves using rows of cilia.
- **Amebas** move by extending **pseudopodia**.
 - **Lobopodia**: Blunt-tipped.
 - **Filopodia**: Thin, sharply pointed.
 - **Rhizopodia**: Branched filaments.
 - **Reticulopodia**: Branched filaments forming a net.
 - **Axopodia**: Thin, pointed, with central axial filament of microtubules.



Figure 5: Ameboid movement. At top and center, the ameba extends a pseudopodium toward a **Pandorina** colony. At bottom, the ameba surrounds the **Pandorina** before engulfing it by phagocytosis.

- Ameboid movement involves extending pseudopodia to surround prey.



Figure 6: Examples of amebas. **Amoeba**, **Diffugia**, and **Arcella** have lobopodia; **Chlamydomphrys** has filopodia; and the foraminiferan **Globigerina** bears reticulopodia.

- Examples of amebas with different pseudopodia: **Amoeba** (lobopodia), **Globigerina** (reticulopodia). Some are **testate** (shelled) like **Arcella** and **Diffugia**.

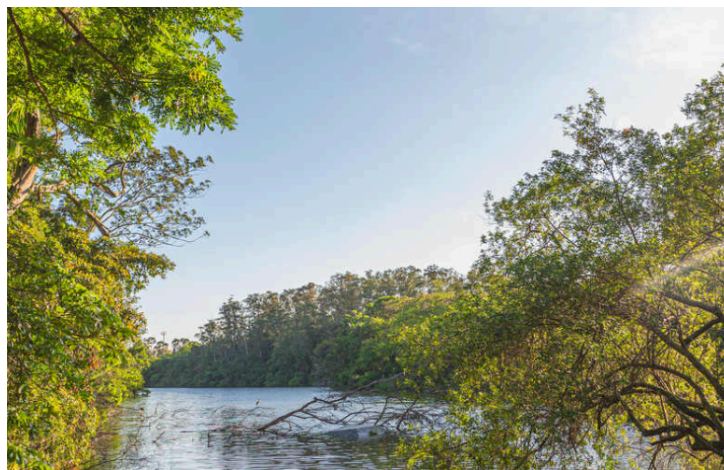


Figure 7: **Actinophrys** and **Clathrulina** are amebas with axopodia.

- **Actinophrys** and **Clathrulina** possess axopodia.



Figure 8: Some shelled amebas, like those shown here, are commonly called radiolarians.

- Radiolarians are shelled amebas.

Cilia and Flagella Structure

- “9 + 2” arrangement of microtubules in the **axoneme**.
- **Kinetosome** (basal body) at the base consists of nine triplets of microtubules (similar to centrioles).
- **Sliding microtubule hypothesis**: Movement powered by ATP and dynein arms causing microtubules to slide past each other.



Figure 9: A, A flagellum illustrating the central axoneme, which is composed of nine pairs of microtubules plus a central pair. The axoneme is enclosed within the cell membrane. The central pair of microtubules ends near the level of the cell surface in a basal plate (axosome). The peripheral microtubules continue inward for a short distance to compose two of each of the triplets in the kinetosome (basal body) (at level y in A). B, Electron micrograph of a section through several cilia, corresponding to level x in A. (133,000x)

- Structure of a flagellum showing the axoneme (9 pairs + 2 central microtubules) and kinetosome.

Pseudopodia Mechanism

- Cytoplasm divided into **ectoplasm** (gel state) and **endoplasm** (sol state).
- Movement involves actin subunits, actin-binding proteins (ABPs), and myosin.
- Hydrostatic force forces endoplasm into the **hyaline cap**.

- Actin polymerizes to form gel-like ectoplasm (tube).
- Contraction at the trailing edge (myosin + actin) forces fluid endoplasm forward.



Figure 10: Ameba in active locomotion. Arrows indicate the direction of streaming endoplasm. The first sign of a new pseudopodium is thickening of the ectoplasm to form a clear hyaline cap, into which the fluid endoplasm flows. As the endoplasm reaches the forward tip, it fountains out and is converted into ectoplasm, forming a stiff outer tube that lengthens as the forward flow continues. Posteriorly the ectoplasm is converted into fluid endoplasm, replenishing the flow.

- Diagram of ameboid movement showing endoplasm streaming into the hyaline cap and conversion to ectoplasm.



Figure 11: A, Electron micrograph of axopodium (from **Actinosphaerium nucleofilum**) in cross section. B, Diagram of axopodium to show orientation of A. The axoneme of an axopodium is composed of an array of microtubules, which may vary from three to many in number depending on the species.

- Cross-section of an axopodium showing the spiral array of microtubules.



Figure 12: Mechanism of pseudopodial movement. In endoplasm, actin subunits are bound to regulatory actin-binding proteins that keep them from assembling (A). Upon stimulation, hydrostatic force carries the subunits through a weakened gel to the hyaline cap. The actin subunits are freed from the regulatory proteins by lipids in the cell membrane (B). Subunits quickly assemble into filaments and, upon interaction with actin-binding protein (ABP), form gel-like ectoplasm (C). At the trailing edge, calcium ions activate an ABP that releases actin filaments from the gel, loosening the network enough that myosin molecules can pull on it (D). Subunits pass up through the tube of ectoplasm to be reused.

- Molecular mechanism of pseudopodial movement involving actin polymerization and myosin interaction.

Functional Components of Protozoan Cells

- **Nucleus:** Membrane-bound, often vesicular. Ciliates have **macronuclei** (compact) and **micronuclei**.
- **Mitochondria:** Energy production. Cristae shape (flat, tubular, discoid, branched) is a taxonomic character.
 - **Hydrogenosomes:** Function in absence of oxygen (mitochondrial derivatives).
 - **Kinetoplasts:** Mitochondrial derivatives with DNA disc.
- **Golgi Apparatus** (dictyosomes): Secretory system. **Parabasal bodies** are similar.
- **Plastids:** Contain photosynthetic pigments (chloroplasts).
- **Extrusomes:** Membrane-bound organelles for extrusion (e.g., **trichocysts**).



Figure 13: Structure of **Acanthamoeba palestinensis**. A, Active, feeding form. B, Cyst.

- Structure of **Acanthamoeba** showing nucleus, mitochondria, and cyst form.

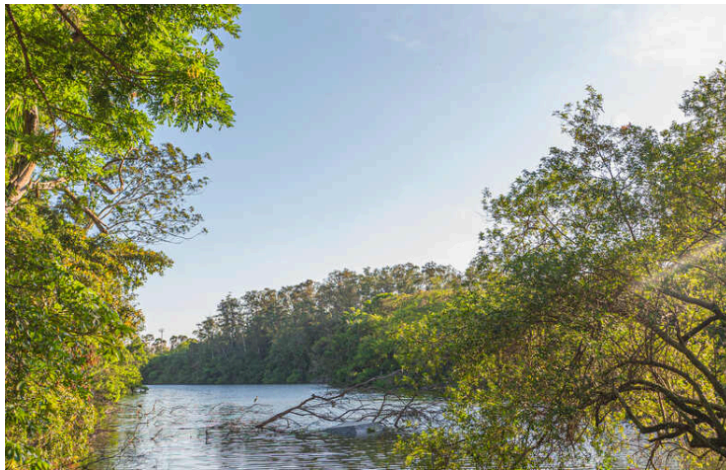


Figure 14: **Euglena viridis**. Features shown are a combination of those visible in living and stained preparations.

- **Euglena viridis** showing chloroplasts, paramylon granules, and flagellum.

Excretion and Osmoregulation

- **Contractile vacuoles**: Function in osmoregulation (removing excess water).
- Common in freshwater forms; rare in marine/parasitic forms.
- Mechanism: Proton pumps transport H^+ and HCO_3^- into vacuole, drawing water in by osmosis.



Figure 15: Left, enlarged section of a contractile vacuole (water expulsion vesicle) of **Paramecium**. Water is apparently collected by endoplasmic reticulum, emptied into feeder canals and then into the vesicle. The vesicle contracts to empty its contents to the outside, thus serving as an osmoregulatory organelle. Right, **Paramecium**, showing cytopharynx, food vacuoles, and nuclei.

- Contractile vacuole of **Paramecium** with feeder canals and ampullae.

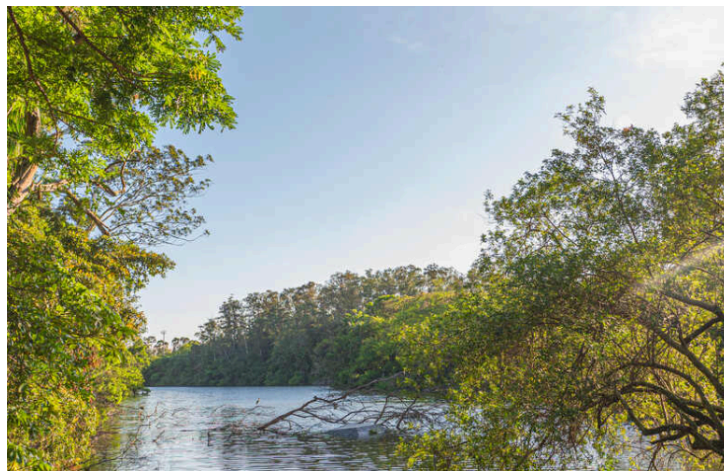


Figure 16: Proposed mechanism for operation of contractile vacuoles. A, B, Vacuoles are composed of a system of cisternae and tubules. Proton pumps in their membranes transport H^+ and cotransport HCO_3^- into the vacuoles. Water diffuses in passively to maintain an osmotic pressure equal to that in the cytoplasm. When the vacuole fills C, its membrane fuses with the cell's surface membrane, expelling water, H^+ , and HCO_3^- . D, Protons and bicarbonate ions are replaced readily by action of carbonic anhydrase on carbon dioxide and water.

- Proposed mechanism: Proton pumps create osmotic gradient to fill vacuole; contents expelled to outside.

Reproduction

- Asexual:
 - **Binary fission**: Two identical individuals.
 - **Budding**: Progeny smaller than parent.
 - **Multiple fission** (schizogony): Several nuclear divisions precede cytokinesis.
- Sexual:
 - **Syngamy**: Fusion of gametes (isogametes or anisogametes).
 - **Conjugation**: Exchange of gametic nuclei between paired organisms.
 - **Autogamy**: Self-fertilization.
- **Encystment**: Formation of dormant **cysts** with resistant coverings to survive harsh conditions.



Figure 17: Binary fission in some amebas and flagellates. A, The two nuclei of **Arcella** divide as some of its cytoplasm is extruded and begins to secrete a new test for the daughter cell. B, The test of another ameba, **Euglypha**, is constructed of secreted platelets. Secretion of platelets for the daughter cell is begun before cytoplasm begins to move out of the aperture.

As these are used to construct the test of the daughter cell, the nucleus divides. C, **Trypanosoma** has a kinetoplast (part of the mitochondrion) near the kinetosome of its flagellum close to its posterior end in the stage shown. All of these parts must be replicated before the cell divides. D, Division of **Euglena**.

- Binary fission in **Arcella**, **Euglypha**, **Trypanosoma**, and **Euglena**.

MAJOR PROTOZOAN TAXA

Opisthokonta

- Flattened mitochondrial cristae; one posterior flagellum.
- Includes metazoans, fungi, **choanoflagellates**, and microsporidians.
- **Choanoflagellates**: Solitary or colonial; likely sister taxon to metazoans.



Figure 18: One hypothesis regarding relationships among some members of Opisthokonta: Choanoflagellates are shown as the sister taxon to Metazoa. Choanoflagellates shown are **Codonosiga** on the left and **Proterospongia** on the right.

- Relationship of choanoflagellates (**Codonosiga**, **Proterospongia**) to Metazoa.

Stramenopiles

- Tubular mitochondrial cristae; heterokont flagellates (two different flagella: one hairy, one smooth).
- Includes brown algae, yellow algae, diatoms, opalinids, and some heliozoans.

Viridiplantae

- Green algae and plants; chloroplasts with chlorophylls **a** and **b**.
- **Phylum Chlorophyta**:
 - **Chlamydomonas**: Single-celled.
 - **Gonium**, **Pandorina**, **Eudorina**: Colonial.
 - **Volvox**: Hollow sphere of thousands of cells; division of labor (somatic vs. reproductive cells).



Figure 19: Examples of phylum Chlorophyta. They are all photoautotrophs.
Chlamydomonas, Gonium, Eudorina, Pandorina.

- Examples of Chlorophyta: **Chlamydomonas, Gonium, Eudorina, Pandorina.**

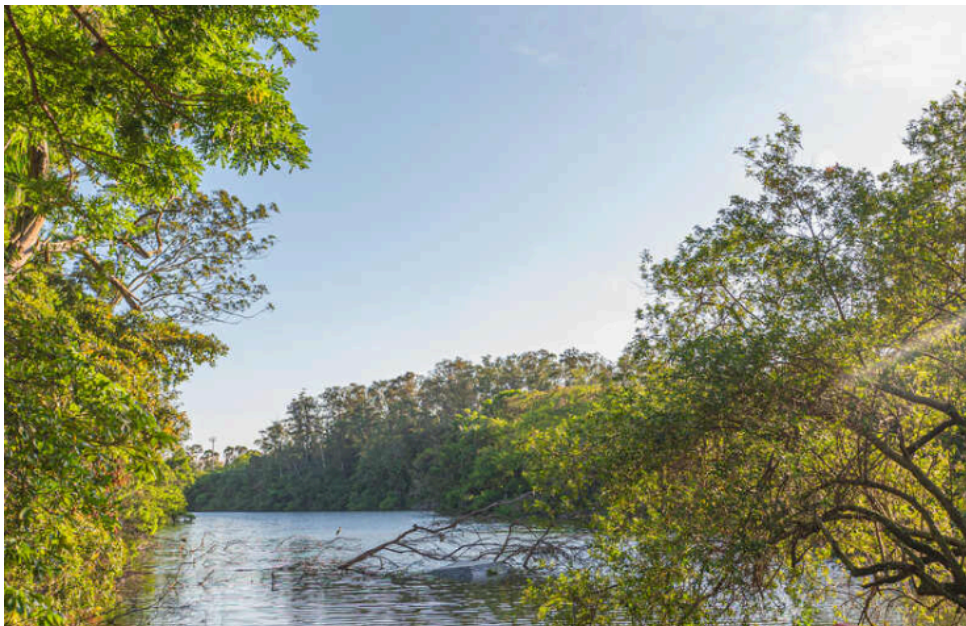


Figure 20: Life cycle of **Volvox**. Asexual reproduction occurs in spring and summer when specialized diploid reproductive cells divide to form young organisms that remain in the mother organism until large enough to escape. Sexual reproduction occurs largely in autumn when haploid sex cells develop. The fertilized ova may encyst and so survive the winter, developing into a mature asexual organism in the spring. In some species the organisms have separate sexes; in others both eggs and sperm are produced in the same organism.

- Life cycle of **Volvox** showing asexual (daughter colonies) and sexual reproduction.

Euglenozoa

- Discoid mitochondrial cristae; longitudinal microtubules beneath cell membrane (**pellicle**).
- **Subphylum Euglenida:**

- Chloroplasts with chlorophyll **b** (secondary endosymbiosis).
- **Euglena**: Flagellum, reservoir, stigma (eyespot), paramylon granules (starchlike storage).
- Nutrition: Autotrophic or saprozoic (if in dark).
- **Subphylum Kinetoplasta**:
 - **Kinetoplast**: Modified mitochondrion with DNA disc.
 - Parasitic.
 - **Trypanosoma**: Causes African sleeping sickness (**T. brucei**, vector: tsetse fly) and Chagas' disease (**T. cruzi**, vector: kissing bug).
 - **Leishmania**: Causes visceral, cutaneous, or mucocutaneous diseases (vector: sand flies).



Figure 21: Examples of phylum Euglenozoa. **Peranema** is a colorless, free-living phagotroph, and **Phacus** is a green, free-living photoautotroph. **Trypanosoma** and **Leishmania** are parasitic, and some species cause serious diseases of humans and domestic animals. **Leishmania** is shown as its intracellular form, without an external flagellum.

- **Peranema, Phacus, Trypanosoma, and Leishmania.**

Retortamonada and Diplomonads

- Lack mitochondria (secondary loss).
- **Diplomonads**: **Giardia lamblia** (intestinal parasite causing diarrhea).



Figure 22: **Giardia lamblia** often causes diarrhea in humans.

- **Giardia lamblia**.

Alveolata

- Shared presence of **alveoli** (membrane-bound sacs beneath cell membrane).
- Includes Ciliophora, Dinoflagellata, and Apicomplexa.

Phylum Ciliophora

- Body covered with **cilia**.
- Multinucleate: **Macronucleus** (metabolic/developmental) and **Micronucleus** (sexual).
- **Pellicle** armor; **infraciliature** (kinetosomes + fibrils).
- **Trichocysts** and **toxicysts** (defensive extrusomes).
- **Cytostome** (mouth) and **cytoproct** (waste expulsion).



Figure 23: Some representative ciliates. **Euplotes** have stiff cirri used for crawling about. Contractile fibrils in ectoplasm of **Stentor** and in stalks of **Vorticella** allow great expansion and contraction. Note the macronuclei, long and curved in **Euplotes** and **Vorticella**, shaped like a string of beads in **Stentor**.

- Representative ciliates: **Euplotes**, **Tetrahymena**, **Dileptus**, **Stentor**, **Vorticella**.



Figure 24: Infraciliature and associated structures in ciliates. A, Structure of the pellicle and its relation to the infraciliature system. B, Expelled trichocyst.

- Infraciliature and trichocysts.
- **Suctorians:** Young are ciliated/free-swimming; adults are sessile with tentacles for feeding (**Podophrya**).
- **Symbiotic Ciliates:** **Balantidium coli** (parasitic in humans), **Ichthyophthirius** (“ick” in fish), **Entodinium** (ruminants).



Figure 25: Some symbiotic ciliates. **Balantidium coli** is a parasite of humans and other mammals. **Ichthyophthirius** causes a common disease in aquarium and wild freshwater fishes. **Entodinium** is found in the rumen of cows and sheep.

- **Ichthyophthirius, Entodinium, Balantidium coli.**
- **Paramecium:**
 - Slipper-shaped; oral groove; pellicle; cytostome; cytopharynx; contractile vacuoles.
 - Holozoic feeder (bacteria, algae).
 - Movement: Spiral path; **avoiding reaction** (reverses cilia upon stimulus).



Figure 26: A, Spiral path of swimming **Paramecium**. B, Avoidance reaction of **Paramecium**.

- Spiral swimming path and avoiding reaction of **Paramecium**.
- Reproduction in **Paramecium**:
 - **Binary fission**: Micronucleus divides mitotically; macronucleus amitotically.
 - **Conjugation**: Exchange of micronuclear material between two individuals (sexual).
 - **Autogamy**: Self-fertilization.

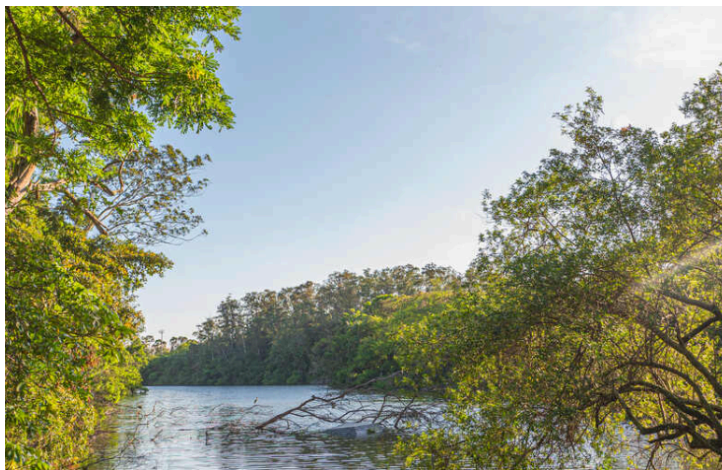


Figure 27: Binary fission in a ciliophoran (**Paramecium**). Division is across rows of cilia.

- Binary fission in **Paramecium**.



Figure 28: Scheme of conjugation in **Paramecium**. A, Two **Paramecium** individuals come into contact on their oral surface. B, The micronuclei divide by meiosis to produce four haploid micronuclei. Macronuclei degenerate. C, Three micronuclei degenerate; the remaining micronucleus divides to form “male” and “female” pronuclei. D, Male pronuclei are exchanged between conjugants. E, Male and female pronuclei fuse to make a diploid nucleus, and individuals separate. F, Three sets of mitotic divisions produce eight micronuclei; four of these become macronuclei while three degenerate. G, The remaining micronucleus divides twice as does the cell, producing four daughter cells.

- Steps of conjugation in **Paramecium**: Meiosis of micronuclei, exchange of pronuclei, fusion to form diploid nucleus.

Phylum Dinoflagellata

- Two flagella (equatorial and longitudinal) in grooves.
- Many photoautotrophic; important primary producers (zooxanthellae in corals).
- Some produce “red tides” and toxins (**Pfiesteria**, **Ciguatera**).
- **Noctiluca**: Bioluminescent, voracious predator.



Figure 29: Examples of phylum Dinoflagellata. **Gymnodinium** bears no cellulose plates. Some members of its family are autotrophic and some phagotrophic. **Ceratium** bears plates and is both autotrophic and phagotrophic. **Noctiluca** is entirely phagotrophic, can be very large (more than 1 mm wide), and has a large tentacle involved in feeding.

- **Ceratium, Noctiluca, Gymnodinium.**

Phylum Apicomplexa

- Endoparasites; **Apical complex** (rhoptries, micronemes, etc.) for penetrating host.
- Life cycle: Asexual (schizogony) and sexual (sporogony); **spore** (oocyst).
- **Class Coccidea:**
 - **Eimeria:** Coccidiosis in fowl.
 - **Toxoplasma gondii:** Toxoplasmosis (cats are definitive hosts); dangerous for pregnant women and AIDS patients.
 - **Plasmodium:** Malaria (Vector: **Anopheles** mosquito).
 - **P. falciparum** (most virulent), **P. vivax**, **P. malariae**, **P. ovale**.
 - Life cycle: Sporozoites injected -> Liver (schizogony) -> Merozoites -> Red blood cells (trophozoites -> schizogony) -> Gametocytes -> Mosquito (sexual cycle -> sporozoites).



Figure 30: A, Diagram of an apicomplexan sporozoite or merozoite at the electron-microscope level, illustrating the apical complex. The polar ring, conoid, micronemes, rhoptries, subpellicular microtubules, and micropore (cytostome) are all considered components of the apical complex. B, Infective oocyst of **Eimeria**. The oocyst is the resistant stage and has undergone multiple fission after zygote formation (sporogony).

- Apical complex structure and **Eimeria** oocyst.

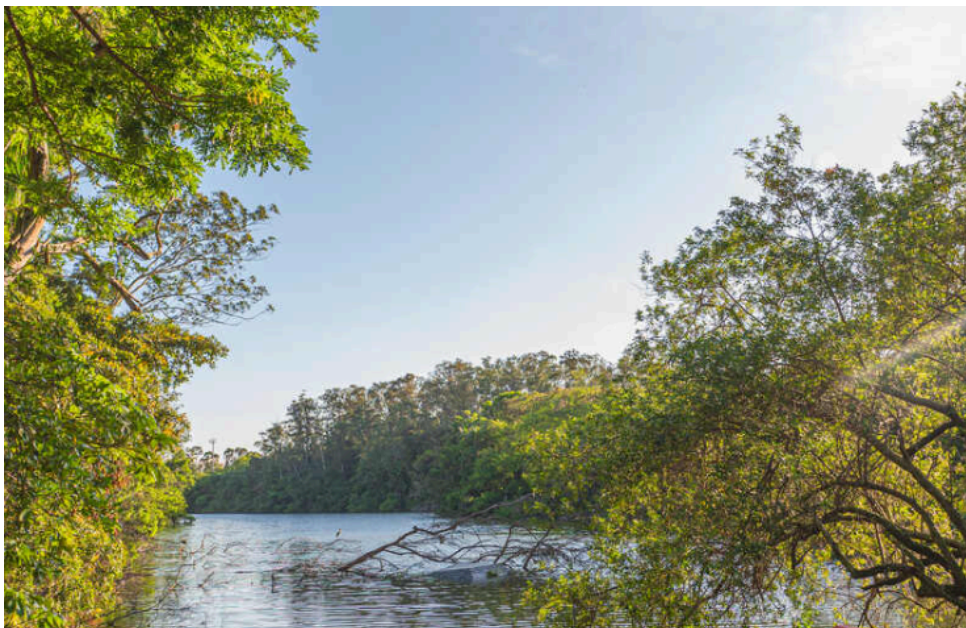


Figure 31: Life cycle of **Plasmodium vivax**, one of the protozoa (class Coccidia) that causes malaria in humans. A, Sexual cycle produces sporozoites in body of mosquito. Meiosis occurs just after zygote formation (zygotic meiosis). B, Sporozoites infect a human and reproduce asexually, first in liver cells and then in red blood cells. Malaria is spread by **Anopheles** mosquito, which ingests gametocytes along with human blood, then, when biting another victim, leaves sporozoites in new wound.

- Life cycle of **Plasmodium vivax**.

Parabasalids

- **Axostyle** (stiffening rod of microtubules).
- **Parabasal body** (modified Golgi).
- **Trichomonas vaginalis**: Causes vaginitis (sexually transmitted).
- Symbionts in termites (**Trichonympha**).



Figure 32: These three animals belong to the parabasalid clade. **Trichomonas vaginalis** is transmitted sexually and is a frequent cause of vaginitis in humans. **Trichonympha** and **Spirotrichonympha** are mutualistic symbionts in termites.

- **Trichomonas, Trichonympha, Spirotrichonympha.**

Amebas

- Polyphyletic group.
- **Nonactinopod Amebas**:
 - **Rhizopodans**: Lobopodia, filopodia, or rhizopodia.
 - **Amoeba proteus**: Freshwater, lobopodia.
 - **Heterolobosea**: **Naegleria fowleri** (amebic meningitis).
 - **Entamoebidae**: **Entamoeba histolytica** (amebic dysentery).
- **Granuloreticulosa**:
 - **Foraminiferans**: Reticulopodia; calcium carbonate tests (many chambered).
 - Important fossils; form limestone/chalk deposits (White Cliffs of Dover).



Figure 33: A, Living foraminiferan, showing thin pseudopodia extending from test. B, Test of foraminiferan, **Vertebralina striata**. Foraminiferans are ameboid marine protozoa that secrete a calcareous, many-chambered test in which to live and then extrude protoplasm through pores to form a layer over the outside. The animal begins with one chamber, and as it grows, it secretes a succession of new and larger chambers, continuing this process throughout life. Many foraminiferans are planktonic, and when they die, their shells are added to the ooze on the ocean's bottom.

- Foraminiferan with reticulopodia and test.
- **Actinopod Amebas:**
 - **Heliozoans:** Freshwater, axopodia (**Actinosphaerium**).
 - **Radiolarians:** Marine, siliceous skeletons, central capsule. Oldest known protozoa.



Figure 34: Types of radiolarian tests. In his study of these beautiful forms collected on the famous Challenger expedition of 1872 to 1876, Haeckel proposed our present concepts of symmetry.

- Radiolarian tests.

PHYLOGENY AND ADAPTIVE DIVERSIFICATION

- Molecular evidence suggests eukaryotic diversification into supergroups (Chromalveolates, Unikonts, Plantae, Rhizaria, Excavates).